AP Calculus BC Unit 10 Review Book

Infinite Sequences and Series

This review book summarizes every topic in Unit 10 in a way that is meant to be practical for final review. The emphasis is on recognition: when you see a series, how do you decide what to do first, what result the test gives, and what traps to avoid?

Think of the unit in three layers. Layer 1: convergence and divergence tests for infinite series. Layer 2: absolute versus conditional convergence and error bounds. Layer 3: Taylor, Maclaurin, and power series for approximation and function representation.

1. Core notation

A sequence is written as \(a_n\). A series is the infinite sum of those terms. A series converges when its partial sums converge.

$$S_n = a_1 + a_2 + \cdots + a_n$$

$$\sum_{n=1}^{\infty} a_n$$

10.1 Convergent and divergent infinite series

- A series converges when the partial sums approach a finite limit.
- A series diverges when the partial sums fail to approach a finite limit.

- Do not confuse the limit of the terms with the sum of the series.

Example

Question. Suppose the partial sums are \(S_n=5-\frac{2}{n}\). Find the sum of the series.

$$S_n = 5 - \frac{2}{n}$$

$$\lim_{n\to\infty} S_n = 5$$

Solution. Take the limit of the partial sums. Since \(\frac{2}{n}\to 0\), the series converges to 5.

10.2 Geometric series

- A geometric series has a constant common ratio.
- It converges if and only if the ratio has absolute value less than 1.
- If it converges, use the closed-form sum immediately.

Rule

Question. Memorize the geometric sum formula.

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$$

$$|r| < 1$$

Solution. This is one of the few families whose sum can be found exactly without approximation.

Example

Question. Determine whether \(\sum_{n=1}^{\infty}\frac{3}{2^n}\) converges, and find the sum.

$$\sum_{n=1}^{\infty} \frac{3}{2^n} = 3$$

Solution. It is geometric with first term \(\frac32\) and ratio \(\frac12\). Since \(|r|<1\), it converges, and the sum is 3.

10.3 The n-th term test for divergence

- If a series converges, then the terms must go to zero.
- If the terms do not go to zero, the series diverges immediately.
- If the terms do go to zero, the test gives no conclusion.

Rule

Question. Memorize the divergence trigger.

$$\lim_{n\to\infty} a_n \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ diverges}$$

Solution. This is usually the first quick check you should make.

10.4 Integral test

- Use the integral test when the terms come from a positive, continuous, decreasing function.
- The series and the corresponding improper integral have the same convergence behavior.
- This test is especially useful with logarithmic denominators.

Example

Question. Determine whether \(\sum_{n=2}^{\infty}\frac{1}{n(\ln n)^2}\) converges.

$$\int_2^{\infty} \frac{1}{x(\ln x)^2}\, dx$$

$$u = \ln x, \qquad du = \frac{1}{x}\, dx$$

Solution. Use the integral test. With the substitution $u=\ln x$, the integral becomes $\int \frac{1}{u^2}\,du$, which converges. Therefore the series converges.

10.5 Harmonic series and p-series

- The harmonic series diverges.
- A p-series converges if and only if $p>1$.
- These are benchmark series for comparison.

Rules

Question. Memorize both benchmark families.

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges iff } p > 1$$

Solution. You will use them again and again in comparison arguments.

10.6 Comparison tests

- Use direct comparison when a simple inequality is obvious.
- Use limit comparison when two series have the same dominant behavior.
- Comparison tests require positive terms.

Example

Question. Determine whether $\sum_{n=1}^{\infty}\frac{1}{n^2+1}$ converges.

$$0 < \frac{1}{n^2+1} < \frac{1}{n^2}$$

Solution. Because $0<\frac{1}{n^2+1}<\frac{1}{n^2}$ and $\sum \frac{1}{n^2}$ converges, the given series converges by direct comparison.

Example

Question. Determine whether $\sum_{n=1}^{\infty}\frac{3n^2+1}{n^3+5}$ converges.

$$\lim_{n\to\infty} \frac{\frac{3n^2+1}{n^3+5}}{\frac{1}{n}} = 3$$

Solution. Compare with $\sum \frac{1}{n}$. The limit of the ratio is 3, so both series have the same behavior. The harmonic series diverges, so the given series diverges.

10.7 Alternating series test

- An alternating series usually has a factor such as $(-1)^n$ or $(-1)^{n+1}$.
- If the positive part decreases to zero, the series converges by the alternating series test.
- This proves convergence of the original series, not absolute convergence.

Rule

Question. Use the alternating series test on the positive part b_n.

$$b_n \downarrow 0$$

Solution. You need two things: decrease and a limit of zero.

10.8 Ratio test

- The ratio test is strongest for factorials, exponentials, and powers such as \(n^n\).
- If the ratio limit is less than 1, the series converges absolutely.
- If the ratio limit is 1, the test gives no conclusion.

Rule

Question. Write the ratio limit carefully.

$$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

Solution. Take absolute value before simplifying.

Example

Question. Determine whether \(\sum_{n=1}^{\infty}\frac{2^n}{n!}\) converges.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2}{n+1} \to 0$$

Solution. The ratio becomes \(\frac{2}{n+1}\), which tends to 0. Therefore the series converges absolutely.

10.9 Absolute and conditional convergence

- Absolute convergence means the absolute-value series converges.

- Conditional convergence means the original series converges but the absolute-value series diverges.
- Always check the absolute-value series if the question asks for a classification.

Example

Question. Classify \(\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}\).

$$\sum_{n=1}^{\infty}\left|\frac{(-1)^{n+1}}{n}\right| = \sum_{n=1}^{\infty}\frac{1}{n}$$

Solution. The original series converges by the alternating series test, but the absolute-value series is harmonic and diverges. Therefore the series is conditionally convergent.

10.10 Alternating series error bound

- For a convergent alternating series, the error after using the first \(n\) terms is at most the next omitted term.
- This gives a quick way to choose how many terms are needed for a desired accuracy.

Rule

Question. Memorize the next-term error estimate.

$$|R_n| = |S - S_n|$$

$$|R_n| \leq b_{n+1}$$

Solution. This is one of the fastest calculations in the chapter.

Example

Question. How many terms of \(\sum_{k=1}^{\infty}(-1)^{k+1}\frac{1}{k}\) are needed to make the error less than \(0.001\)?

$$\frac{1}{n+1} < 0.001$$

Solution. Require \(\frac{1}{n+1}<0.001\). This gives \(n+1>1000\), so at least 1000 terms are needed.

10.11 Taylor polynomial approximations

- A Taylor polynomial approximates a function near a chosen center.
- A Maclaurin polynomial is centered at 0.
- The degree is the highest power retained.

Rule

Question. Write the Taylor polynomial in standard form.

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

Solution. The formula below is the finite polynomial version.

Example

Question. Find the third-degree Maclaurin polynomial for \(e^x\).

$$P_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

Solution. Every derivative of \(e^x\) equals \(e^x\), so all derivative values at 0 are 1. Therefore the polynomial is \(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}\).

10.12 Lagrange error bound

- The Lagrange bound estimates the maximum possible error after truncating a Taylor polynomial.
- For a degree-\(n\) polynomial, use a bound on the \((n+1)\)-st derivative over the interval.

Rule

Question. Use the standard error-bound formula.

$$|R_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}$$

Solution. Remember that this is an upper bound, not the exact error.

10.13 Radius and interval of convergence of power series

- A power series is usually centered at a point \(a\).
- The ratio test often gives the radius of convergence.
- You must still test the endpoints separately to get the interval of convergence.

Rule

Question. General form of a power series.

$$\sum_{n=0}^{\infty} c_n(x-a)^n$$

Solution. The center is the value \(a\).

Example

Question. Find the interval of convergence of \(\sum_{n=1}^{\infty}\frac{x^n}{n}\).

$$R = 1$$

$$[-1, 1)$$

Solution. The ratio test gives \(|x|<1\), so the radius is 1. At \(x=1\), the series is harmonic and diverges. At \(x=-1\), it is alternating harmonic and converges. Therefore the interval is \([-1,1)\).

10.14 Taylor and Maclaurin series

- You should memorize the standard Maclaurin series.
- Most AP BC series manipulations come from these templates.
- Substitution, differentiation, and integration create new series from known ones.

Standard series

Question. Memorize the six series below.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$$

Solution. These are the core templates.

10.15 Representing functions as power series

- Build new power series from old ones by substitution, multiplication, differentiation, and integration.
- The geometric series is the most common parent series.
- Check the interval of convergence whenever the problem asks for it.

Example

Question. Represent \(\frac{x}{1-x^2}\) as a power series.

$$\frac{x}{1-x^2} = \sum_{n=0}^{\infty} x^{2n+1}$$

Solution. Start from \(\frac{1}{1-x^2}=1+x^2+x^4+\cdots\), then multiply by \(x\).

Example

Question. Represent \(\frac{1}{(1-x)^2}\) as a power series.

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n$$

Solution. Differentiate the geometric series term by term.

Final strategy map

- Look for geometric structure first.
- Use the n-th term test early to eliminate obvious divergence.
- For positive-term series, think about p-series, comparison, integral test, and ratio test if factorials or exponentials appear.
- For alternating series, decide convergence first, then classify as absolute or conditional if needed.
- For power series, the ratio test is usually the first move and endpoint testing is the last move.